Your first robot factory

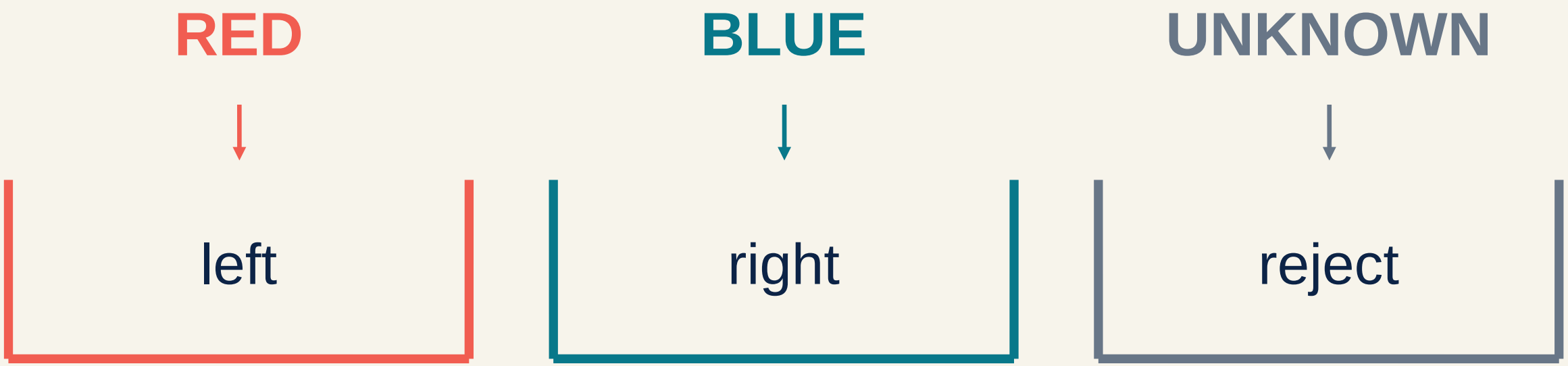
You write the decisions.

What must it know?



Concept illustration

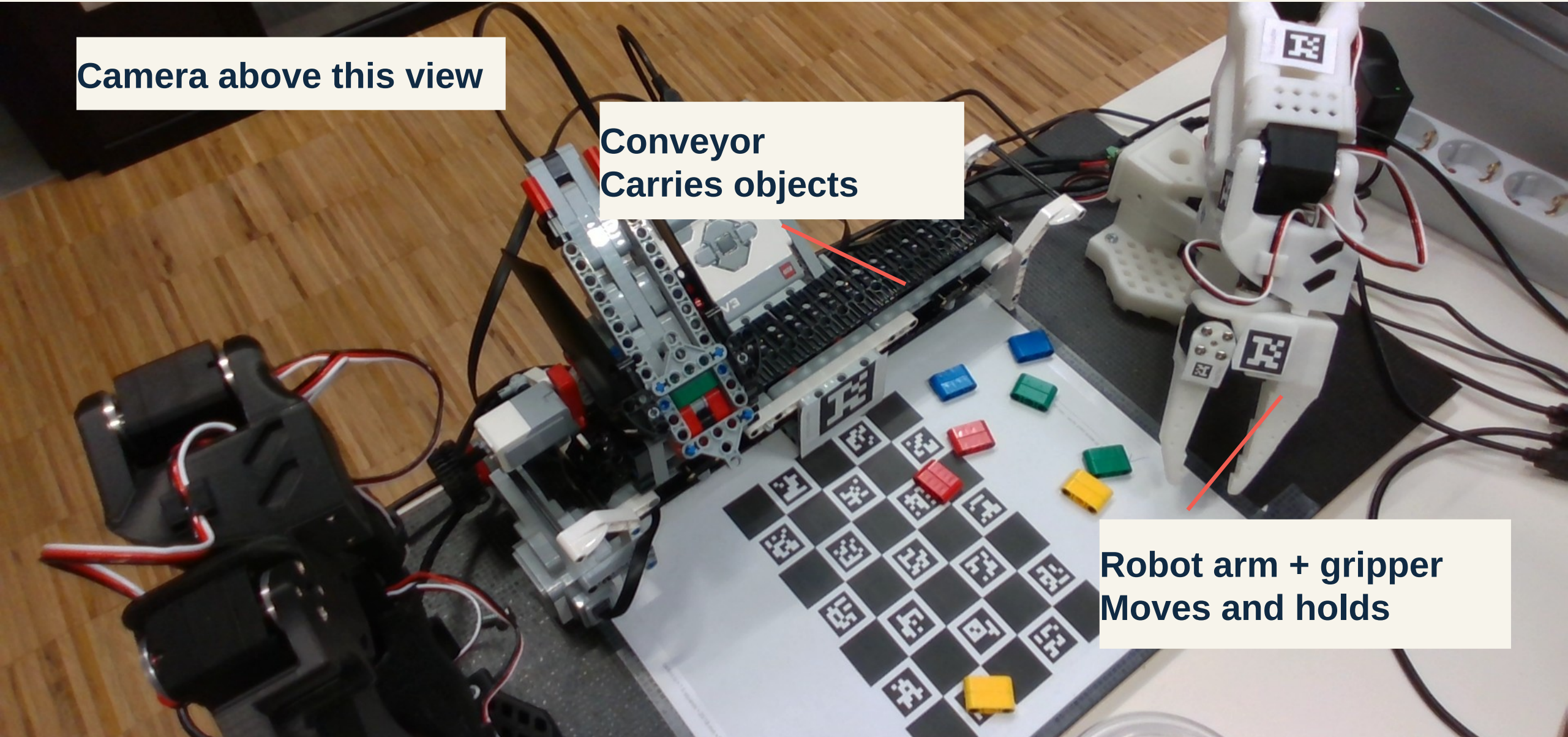
Where does it go?



Predict before we write the rule.

What should happen to a green object?

The equipment in our room



A complete cycle

Observe — Get a picture

↓

Decide — Choose a destination

↓

Act — Make a movement

↓

Check — Did it arrive?



Which step tells us whether it worked?

Information and movement

Sensor

measures

camera → **image**

Actuator

moves

motor → **turn**

Which one can tell us about the color?

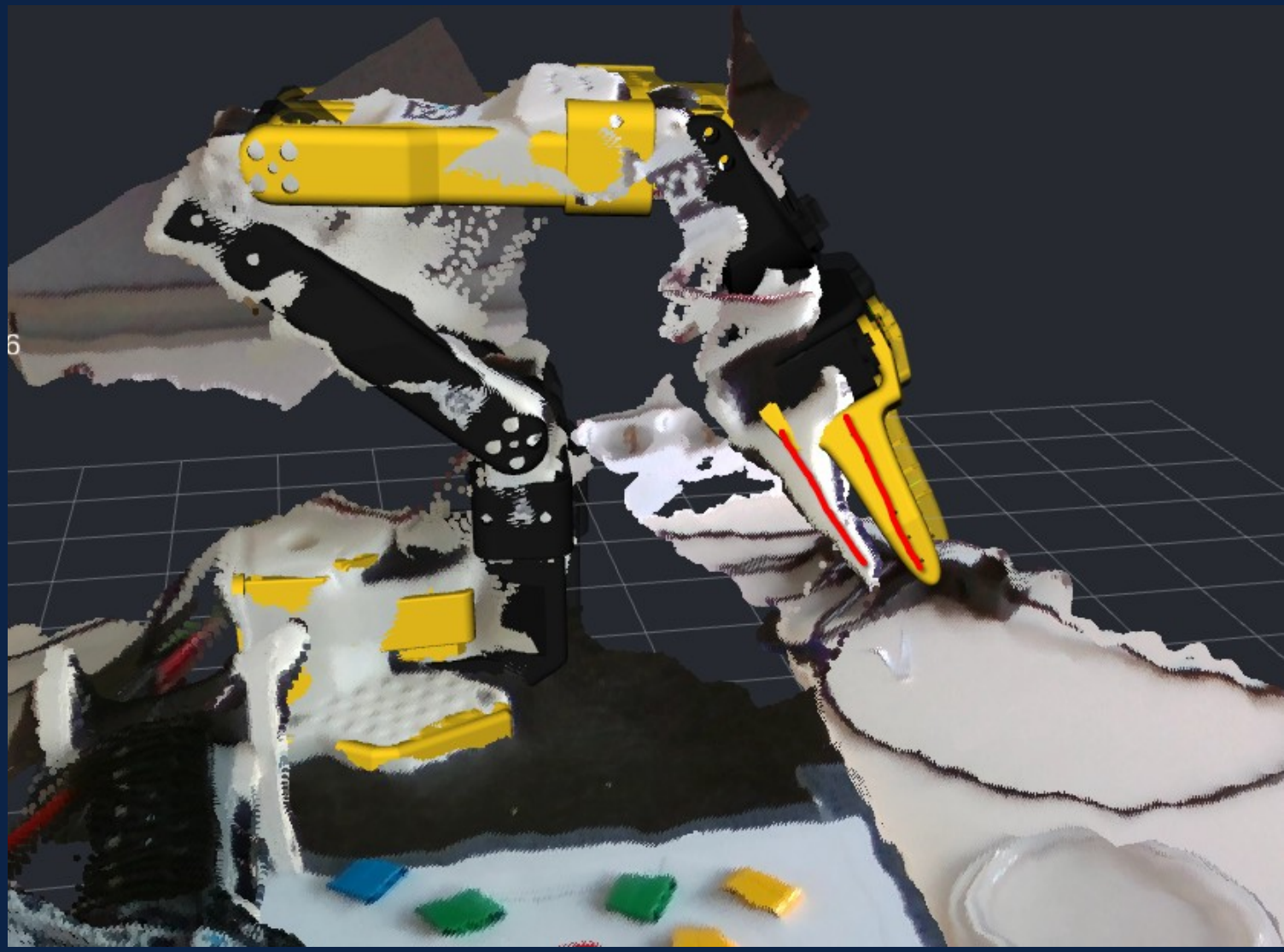
You move.
It follows.

Leader: your handle

Follower: robot arm

Twin: live screen model

A joint bends or rotates.
The gripper opens and closes.



Actual station screen

What does the camera give us?

Picture → **Pixels** → **Label**





“red”

Pixels are tiny squares of color.

Write a rule

red → left

blue → right

We write the choice.

Teach examples

red examples

blue examples

empty background

We show the patterns.

A rule the computer can run

A name stores a value.

COLOR stores text.

Quotes mean text.

if asks a question.

```
COLOR = "red"
if COLOR == "red":
    print("left")
elif COLOR == "blue":
    print("right")
else:
    print("reject")
```

Change red to blue. What will appear?

No matching color

REJECT

Unknown or missing input
uses the reject destination.

Check the result.

Unexpected movement

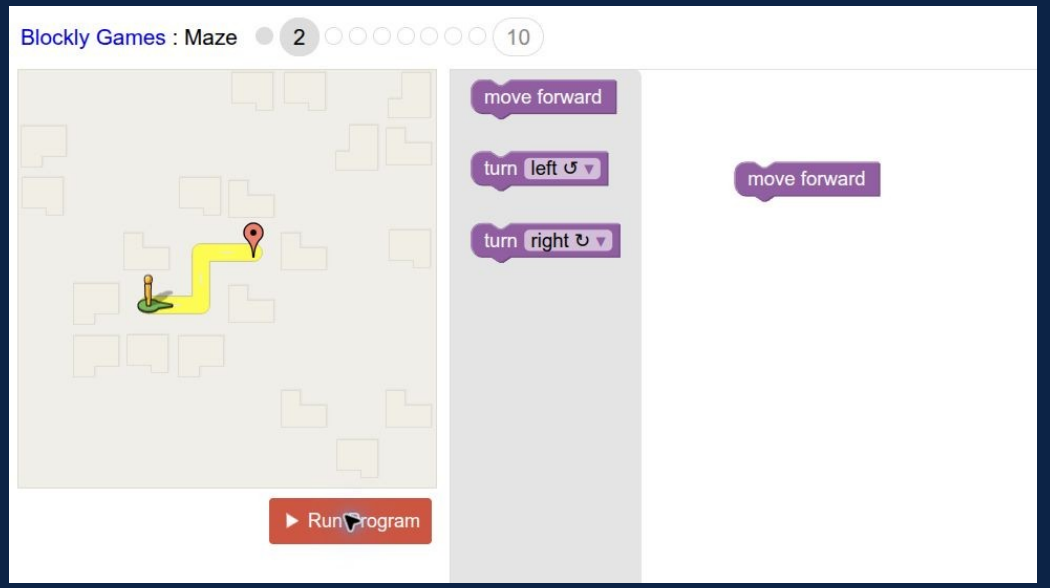
STOP

Ask the instructor
before another physical action.

10 minutes

Build a path

Predict. Build. Run.



Levels 1–3 first.
4–5 if time remains.

Finish: explain one correction.

30 minutes

Change one thing

Predict the output.

Run the cell.

Change the value.

```
COLOR = "blue"
```

?

Finish: a result you predicted correctly.

20 minutes

Teach three answers

red **blue** **background**

Examples → Train → A new view

Finish: explain one mistake or limitation.

A real break

Return at minute 95

10

minutes

Leave station controls with the instructor.

45 minutes

Three stations

1 **Robot + digital twin**

Compare a real joint with the screen.

2 **Camera + conveyor**

Find one reliable reading.

3 **Virtual robot**

Predict one joint movement.

Every 15 minutes, move to the next station.

50 minutes

Make the whole factory work

5

red, blue, red, unknown, blue

2 left + 2 right + 1 reject

Test in simulation first.
Finish: five recorded outcomes.

35 minutes

Choose an upgrade

Swap red and blue destinations.

Count the results.

Stretch: a new color or a bad reading.

Finish: show a test that caught a mistake.

15 minutes

Show your factory

Rule.
Prediction.
Result.

Show one change.
Explain one limitation.
Swap who speaks.

What would you improve next?